Automated Financial Commentary Generation from Structured Financial Data Using Transformer-Based NLP Models

Abstract

Financial analysts routinely transform structured market data, including price movements, trading volume, and volatility measures, into coherent narrative reports that summarize market trends and provide contextual insights. Automating this process remains challenging because it requires accurate numerical reasoning, factual grounding, and domain-specific language generation. In this project, we investigate whether modern transformer-based encoder-decoder models can generate high-quality financial commentary from structured tabular data that closely resembles analyst-written reports.

We formulate the problem as a table-to-text generation task and fine-tune transformer models on the DataTales benchmark, which contains approximately 4,900 pairs of structured financial market data and corresponding expert-written commentaries. We first linearize structured market tables into textual representations capturing price statistics, trading volume, volatility measures, and trend indicators, which are then used as encoder inputs. We fine-tune T5 and BART models on paired input-output examples and evaluate generation quality using both lexical and semantic metrics, including ROUGE, BLEU, METEOR, BERTScore, and Semantic Textual Similarity (STS). Our results demonstrate that while fine-tuned models substantially outperform zero-shot baselines, the modified KAHAN baseline achieves the strongest overall performance across multiple evaluation metrics, outperforming both fine-tuned T5 and BART, with BERTScore F1 = [value] and ROUGE-L = [value]. We further analyze factual consistency using human evaluation via LLM-as-a-judge. Our results show that while fine-tuned models produce fluent and largely factual reports, they still struggle with numerical precision in complex multi-contract scenarios, highlighting opportunities for improved numerical reasoning in data-to-text generation.

1. Introduction

Financial commentary plays a vital role in investment decision-making, yet producing such reports is labor-intensive and requires synthesizing large volumes of structured data into coherent narratives. Financial analysts must identify salient patterns, compute changes over time, and articulate insights in domain-appropriate language. Automating this workflow has significant practical value for financial institutions, research firms, and retail investors, yet remains an open challenge in natural language generation. Models must handle multiple related entities, incorporate specialized terminology, and maintain strict factual fidelity to numerical values. Despite recent advances in natural language generation (NLG), generating financial commentary remains particularly challenging. Financial reports must faithfully represent numerical information while reasoning over multiple related entities, temporal trends, and domain-specific terminology. Transformer-based encoder-decoder architectures have recently achieved state-of-the-art performance on a variety of data-to-text generation tasks, including sports reporting, weather forecasting, and biomedical summarization. Encoder-Decoder architectures have demonstrated strong capabilities in conditional text generation through large-scale pretraining followed by supervised fine-tuning. However, comparatively little work has evaluated these models on financial table-to-text generation, where the combination of structured numerical inputs and domain-specific language introduces additional challenges, including numerical reasoning, multi-contract analysis, and strict factual grounding.

In this project, we investigate the effectiveness of transformer-based encoder-decoder models for automated financial commentary generation using the DataTales benchmark dataset. Specifically, we address the following research questions:

RQ1: Can fine-tuned encoder-decoder transformer models outperform zero-shot and KAHAN baselines on financial data-to-text generation?

RQ2: To what extent do generated reports maintain factual consistency with the underlying numerical data?

RQ3: How do different encoder-decoder architectures compare in terms of lexical quality, semantic similarity, and factual accuracy?

To answer these questions, we develop an end-to-end table-to-text generation pipeline that converts structured financial tables into linearized textual representations before fine-tuning T5 and BART models on paired financial commentary examples. We compare the resulting models against zero-shot prompting and a modified KAHAN rule-based baseline using a comprehensive evaluation framework consisting of lexical metrics (ROUGE, BLEU, and METEOR), semantic metrics (BERTScore and Semantic Textual Similarity),and qualitative assessment through an LLM-as-a-judge framework.

Our experimental results show KAHAN baseline outperforms the fine-tuned transformer models and achieves the strongest lexical and semantic performance while producing fluent, analyst-style financial narratives.

Nevertheless, our analysis reveals persistent limitations in numerical reasoning and factual precision, especially when generating commentary involving multiple financial contracts or complex temporal relationships. These findings demonstrate the promise of transformer-based approaches for financial data-to-text generation while highlighting important directions for future research on improving numerical reasoning and factual grounding in domain-specific natural language generation systems.

2. Background and Related Work

2.1 Data-to-Text Generation

Data-to-text generation is a subfield of natural language generation that aims to transform structured data, such as tables, databases, or knowledge graphs, into coherent natural language descriptions. Early neural approaches demonstrated that encoder-decoder architectures could generate fluent narratives from structured inputs, but often struggled to preserve factual accuracy and numerical consistency. Wiseman et al. (2017) established foundational benchmarks in this area with the introduction of the RotoWire dataset, which pairs NBA box-score statistics with human-written game summaries. Their work highlighted key challenges: while neural models produced fluent text, they often failed to match the quality of human-written documents, and even templated baselines outperformed neural models on certain metrics. Subsequent research has explored copy mechanisms, reconstruction-based extensions, and fact-grounded evaluation to address these limitations. Since then, transformer-based encoder-decoder architectures have become the dominant models for conditional text generation because of their ability to learn complex relationships between structured inputs and natural language outputs through large-scale pretraining and supervised fine-tuning.

2.2 Financial Table-to-Text Generation

The financial domain presents distinct challenges for data-to-text generation, including specialized terminology, multi-entity reasoning, and strict requirements for numerical accuracy. Chapman et al. (2021) implemented a transformer network for generating financial report text from tables, achieving promising BLEU scores and demonstrating that transformer-based models can produce natural, grammatically correct, and largely faithful sentences. However, their work also revealed that factual consistency, particularly with numerical values, remains a significant challenge.

The DataTales benchmark, introduced by Yang et al. (2024), addresses these challenges by providing 4.9k financial market reports paired with corresponding tabular data. Unlike earlier datasets that primarily require descriptive summaries, DataTales emphasizes higher-level financial reasoning by requiring models to identify trends, explain market movements, compare financial instruments, and generate analyst-style commentary using specialized financial terminology. The benchmark therefore provides a realistic evaluation setting for automated financial reporting and has become a standard dataset for studying financial table-to-text generation.

2.3 KAHAN: Knowledge-Augmented Hierarchical Analysis

Building on the DataTales benchmark, Yang et al. (2025) proposed KAHAN, a framework that performs hierarchical analysis of structured financial data before generating commentary. Rather than relying solely on end-to-end sequence generation, KAHAN explicitly extracts information at multiple levels, including individual entities, pairwise relationships, group interactions, and overall market trends. KAHAN leverages LLMs as domain experts to drive analysis and achieves strong performance on the DataTales benchmark, outperforming existing approaches by over 20% on narrative quality while maintaining 98.2% factuality.

Our work draws inspiration from KAHAN's hierarchical analysis approach. In contrast, we adopt a simpler architecture and supervised learning approach based on encoder- decoder transformers. Rather than using a multi-agent reasoning framework, we evaluate whether fine-tuning T5 and BART models on paired financial data and commentary can achieve competitive performance while requiring substantially less architectural complexity, allowing us to systematically evaluate the effectiveness of transformer-based models on this task.

2.4 Evaluation of Data-to-Text Generation

Evaluating data-to-text generation requires metrics that capture multiple dimensions of quality, including lexical similarity, semantic equivalence, and factual correctness. Most prior work reports lexical overlap metrics such as ROUGE, BLEU, and METEOR, which compare generated text with human-written references. Although these metrics remain widely used, they primarily measure surface-level similarity and may not adequately capture semantic equivalence or factual correctness. Recent studies therefore complement lexical metrics with embedding-based measures such as BERTScore, which evaluates semantic similarity using contextual language representations.

To obtain a more comprehensive assessment of generation quality, recent work has also adopted LLM-as-a-judge frameworks, in which large language models evaluate generated reports on criteria such as factual accuracy, coherence, completeness, and overall readability. These approaches complement automatic metrics by providing qualitative assessments that better reflect human judgment.

Despite advances in automatic evaluation, human evaluation remains the gold standard for assessing data-to-text generation systems. While costly and time-consuming to conduct, human annotators are better equipped to detect subtle factual errors, assess the stylistic appropriateness of generated text, and evaluate the overall narrative coherence and usefulness of generated reports. Many prior works in financial text generation (Chapman et al., 2021; Yang et al., 2025) complement automatic metrics with human evaluation to validate their findings and ensure that improvements measured by automatic metrics correspond to genuinely better outputs from a reader's perspective. In this work, we follow this established practice by employing LLM-as-a-judge as a scalable proxy for human evaluation, while noting that future work should incorporate direct human assessment to further validate our findings.

2.5 Position in the Literature

Our work builds upon prior research in financial data-to-text generation while focusing on the effectiveness of encoder-decoder transformer models. Using the DataTales benchmark, we systematically compare zero-shot and fine-tuned versions of T5 and BART for generating financial commentary from structured market data. Unlike KAHAN, which relies on hierarchical reasoning and large language model orchestration, our approach adopts a simpler supervised learning framework based on sequence-to- sequence transformers. We further provide a comprehensive evaluation combining lexical metrics (ROUGE, BLEU, and METEOR), semantic similarity (BERTScore), and qualitative assessment using an LLM-as-a-judge framework. Through this analysis, we examine both the strengths and limitations of transformer-based financial commentary generation, identifying numerical reasoning as a key area for future improvement.

3. Methodology

3.1 Dataset: DataTales

We use DataTales, a paired dataset of structured market tables and human-written financial commentary reports spanning 11 markets (cattle, corn, currency, dairy, equity market, gold, lean hog, oil, soybean, treasury, wheat). Each example pairs a table of trading statistics (open/close, high/low, volume, price change, volatility) with a corresponding narrative report.

To understand the task and prepare the data for modeling, we conducted a comprehensive EDA of the DataTales dataset, including text statistics and linguistic features. We analyzed the length and structure of the financial reports to understand typical narrative formats, and conducted entity and temporal analysis, examining how reports discuss a subset of available financial instruments (average of 5.69 out of 8.7 entities).

- Total examples: 3,938
- Train / Validation / Test split: 2,627 / 516 / 795 (temporal split per market, validation and test dates strictly follow training dates within each market, confirmed no leakage across all 11 markets)
- Encoder input length: mean 174 tokens, well under the 512-token limit after our compact linearization of the table
- Decoder target length: mean 131 tokens, 90th percentile 257 tokens, 1.3% of targets exceed 384 tokens (max target length used for training)

Table linearization: We convert the raw tabular data into a structured text format by serializing the market data table, where each row represents a financial entity with its values OHLCV over the relevant lookback period. This is similar to the representation used in KAHAN, but without its multi-level hierarchical analysis. Serialization is necessary because pretrained encoder-decoder models (T5, BART) were pretrained on token sequences with embeddings and positional encodings built for linear text, not tabular structure; the encoder requires a flat sequence of tokens as input, so cell/row/column structure cannot be passed directly. Fine-tuning then maps these serialized (input, output) pairs directly to narrative text. A key early finding: raw table strings, before linearization, far exceed typical encoder token limits and compact market summary linearization with period statistics + trend direction was necessary to keep inputs under 200 tokens.

3.2 Model architecture: T5

Our first model is T5 (Text-to-Text Transfer Transformer), fine-tuned for the data narration task. T5's encoder-decoder framework is well suited to generating coherent long-form text from structured input. We fine-tune two variants, T5-small (60M parameters) and T5-base (220M parameters), under matched training conditions to isolate the effect of model scale.

3.3 Model architecture: BART

Our second model is BART that stands for Bidirectional Auto-Regressive Transformers. It is a denoising autoencoder for pretraining sequence-to-sequence models. Facebook/bart-base model has 139.4M trainable parameters.

3.3.1 BART – Training setup

BART architecture uses tokenizer_bart to tokenize both inputs and targets, ensuring they are truncated to Max_input_length (1024) and Max_target_length (1024), respectively. A higher threshold is used to get the least truncation of input and output. This is resulting in 0% input and output exceeding this threshold. The following setup was used for training:

- eval_strategy= 'epoch': Evaluation metrics are calculated at the end of each training epoch.
- learning_rate=5e-5: The rate at which the model's weights are updated.
- per_device_train_batch_size=8: Number of training examples processed at once per GPU/CPU.
- num_train_epochs=5: The total number of passes over the entire training dataset. A trial was done with 10 epoch but it tends towards overfitting therefore 5 epochs were used.
- predict_with_generate=True: Ensures the model generates text during evaluation, which is crucial for sequence-to-sequence tasks.
- fp16=True: Enables mixed-precision training (if a compatible GPU is available) for faster training.

After training, the model's performance is further evaluated using various metrics (ROUGE, SacreBLEU, BERTScore, METEOR) and a K-Fold cross-validation strategy on the test set, demonstrating a comprehensive fine-tuning and evaluation workflow.

3.4 T5- Training setup

Both T5 variants share the same encoder input template and decoder target, dynamic padding via DataCollatorForSeq2Seq, and EarlyStoppingCallback(patience=2). Validation loss was monitored on a leakage-safe validation split carved from the training dates (not the test set). T5-small trained for 10 epochs at learning rate 3e-4 (effective batch size 8); T5-base trained with the same learning rate and effective batch size (per-device batch 4, gradient accumulation 2) but was early-stopped at 6 epoch.

3.5 Baseline: KAHAN

The baseline model is adapted from the KAHAN paper and fit for this use case. The KAHAN model works as a set of LLM instructions as follows:
First, the LLM is prompted to propose 5 analysis questions about this entity and what metrics to compute.
Then, it is tasked to write Python code that computes those metrics from this data and gets back plain numeric output.
Next, the LLM is tasked to find the significance behind the numbers that were received and any relationships or patterns that they may have.
Finally, with this additional context, the LLM can generate a cohesive report given historical data and patterns

3.6 Evaluation metrics

Lexical overlap: ROUGE-1/2/L, sacreBLEU
Semantic similarity: METEOR, BERTScore (P/R/F1), Sentence-BERT cosine similarity (all-MiniLM-L6-v2)
Stability: post-hoc 5-fold analysis on held-out test predictions (mean ± std across folds; not true k-fold CV since models were trained once rather than retrained per fold this measures metric stability, not training variance)
Uncertainty: bootstrap 95% CI (2,000 resamples) on STS scores

Factual grounding (custom diagnostic): numeric support rate fraction of numbers in generated text that approximately match values present in the source table
LLM-as-a-judge: GPT-4o-mini rates holistic quality (1–10 scale) on fluency, coherence, and factual alignment with the reference, sampled over 50 test examples per model

4. Results And Discussion

4.1 Final Test-Set Performance

Table 1 reports performance on the held-out test set of 795 examples. The temporal train-validation-test split ensures that test observations occur after the corresponding training observations within each market, reducing the risk of temporal leakage.

Fine-tuning substantially improved both T5 models relative to their zero-shot baselines. Across the fine-tuned T5 models, T5-small achieved the best performance on both lexical and semantic metrics: sacreBLEU (5.49 vs. 1.43) and SBERT similarity (0.636 vs. 0.556), while also outperforming T5-base on ROUGE (0.26 vs. 0.17), and BERTScore (0.86 vs. 0.83). The results suggest that increasing model size did not improve performance under the training configuration used in this study. This also highlights the value of supervised fine-tuning[2] for learning the domain-specific vocabulary and reporting patterns required for financial table-to-text generation.

The fine-tuned BART model achieved moderate lexical overlap with the reference reports. On the held-out test set, the model obtained a ROUGE-1 score of 0.2601, ROUGE-2 of 0.0894, and ROUGE-L of 0.1772. The model also achieved a sacreBLEU score of 5.4026 and a METEOR score of 0.1726, indicating moderate n-gram overlap while accounting for stemming and synonymy.

In terms of semantic performance, the model achieved a BERTScore-F1 of 0.8765, indicating strong semantic similarity between the generated and reference reports despite the moderate lexical overlap. Overall, these results suggest that the fine-tuned BART model is able to capture the underlying financial concepts while producing reports that differ lexically from the reference text.

4.2 Post-hoc Test-Set Stability

To determine whether the observed model differences were driven by particular subsets of the test data, the 795 held-out predictions were partitioned into five folds of 159 examples each and evaluation metrics were recomputed independently for each fold. This is a post-hoc stability analysis rather than conventional k-fold cross-validation because the models were trained only once.

4.2.1 T-5 Post-hoc Stability: Table 3

Both T5-small and T5-base show relatively low variation across test folds, indicating that their aggregate performance is not driven by a small number of strong or weak examples. T5-small also outperforms T5-base consistently across the partitions.

4.2.2 BART Post-hoc Stability Table 5

The BART model exhibits relatively low variation across the five test partitions, indicating that its aggregate performance is not driven by a small number of particularly strong or weak examples. Low standard deviations across ROUGE-L, sacreBLEU, BERTScore-F1, and METEOR suggest that the model performs consistently across different subsets of the held-out test set. These results indicate that the reported test-set performance is stable and robust, providing greater confidence that the model will generalize reliably to unseen data drawn from the same distribution.

4.3 Numerical Grounding: Table 4
Lexical and semantic similarity do not directly establish whether generated financial values are supported by the underlying data. We therefore computed a numeric support diagnostic measuring the proportion of numbers generated in each commentary that approximately match numerical values available in the corresponding model input. Both T5 models scored low as only less than half of the generated numeric values can be matched to values in the structured input. This represents a substantial limitation for automated financial commentary, where numerical fidelity is critical. The result also illustrates why lexical and semantic metrics are insufficient for evaluating financial data-to-text systems. Generated reports can resemble the reference linguistically while

To further evaluate semantic alignment, we computed a Semantic Textual Similarity (STS) score using DistilGPT2 embeddings. The fine-tuned BART model achieved an STS score of **0.9160**, indicating a high degree of semantic similarity between the generated and reference reports. This result suggests that the model is able to capture the underlying financial information and reporting patterns while producing semantically consistent commentaries. However, as with other semantic similarity metrics, a high STS score does not necessarily guarantee factual correctness or numerical grounding, as semantically similar reports may still contain unsupported or inaccurate numerical values.

4.4 LLM-as-a-Judge Evaluation
Finally, we complement the automatic metrics with an LLM-as-a-judge evaluation using GPT-4o-mini on a sample of 50 test examples per model. The judge assigns an overall quality score from 1-10 based on fluency, coherence, and factual alignment with the reference. A separate LLM-based decoder similarity score measures semantic correspondence on a 0-1 scale.

Model	K-Fold + STS	LLM-as-a-judge
Baseline(KAHAN)	5.46 ± 0.39	3.30 ± 0.70
NM-Bart	0.1725 ± 0.0453	2.94 ± 0.90
T5-small (fine-tuned)	0.130 ± 0.132	2.72 ± 0.80
T5-base (fine-tuned)	0.035 ± 0.087	1.72 ± 0.72

The absolute LLM-judge scores are low for both models. This is consistent with the numerical-grounding analysis, which identified substantial unsupported numerical generation. The baseline KAHAN model did the best by far for the semantic and lexical metrics, and was still slightly better with the LLM-as-a-judge score, indicating that it is still the better model for the task.

5. Limitations and Next Steps

While newer models like GPT-4 and LLaMA-2 achieve state-of-the-art results, we selected T5 and BART due to their computational manageability, established architectures and open-source accessibility for reproducible research.

We use standard supervised fine-tuning without domain-adaptive pre-training (DAPT) or parameter-efficient fine-tuning (PEFT). Future work could explore DAPT on financial corpora and (PEFT) methods such as LoRA, enabling performance improvement, reduced computational cost and allow for larger model experimentation. The T5 tokenizer's subword segmentation led to limited numerical reasoning (see Wallace et al. 2019); future work could explore number-specific tokenization orre-processing such as z-score normalization of numerical features. Another promising direction is hierarchical input construction method that aggregates raw tabular data into structured summaries capturing price statistics, volatility, volume, and trend information. Finally future work could compare encoder- decoder baselines with decoder-only LLMs (e.g., instruction-tuned models) for table-to-text generation.

6. Conclusion

This project demonstrates that the T5 and BART models can parallel the KAHAN model on lexical and semantic measures such as ROUGE, BLEU, and BERTScore with the T5 being the best fine tuned model. However when grading holistically, the KAHAN baseline outperformed the transformers, meaning that these metrics do not always fully capture the qualities like factual grounding and coherent reasoning. This makes KAHAN a more trustworthy model due to its structured reasoning and thinking process.

Author Contributions:

Sid: Dataset generation, KAHAN model, LLM-as-a-judge evaluation

Priyanka: Did the project ideation, literature review, identified dataset. Implemented and evaluated T5-small and T5-base models. Contributed to the project report’s sections including Abstract, Introduction, Limitations and Next steps, Methodology and Related work for T5.

Niha: BART model, experimented with importing HuggingFace models, and did STS experimentation, results and discussion for BART model

Appendix

1. **Table 1**

Model	ROUGE-1	ROUGE-2	ROUGE-L	sacreBLEU	METEOR	BERTScore F1	SBERT STS
T5-small (zero-shot)	0.058	0.001	0.041	0.004	0.026	0.776	—
T5-small (fine-tuned)	**0.259**	**0.087**	**0.191**	**5.49**	**0.162**	**0.856**	**0.636**
T5-base (zero-shot)	0.058	0.004	0.040	0.224	0.033	0.765	—
T5-base (fine-tuned)	0.172	0.050	0.137	1.43	0.088	0.829	0.556
BART	0.2693	0.0898	0.1772	5.4026	0.1726	0.8591	0.9911

2. **T5-base Model Scale and Training Dynamics**

The poorer performance of T5-base was unexpected because it contains approximately 220M parameters compared with 60M for T5-small. Examination of the training dynamics, however, suggests that the larger model did not converge as effectively under the shared training configuration.

T5-small completed 10 epochs with a final training loss of 3.11. In contrast, T5-base was early-stopped after six epochs with a final training loss of 7.73. Its validation loss decreased from 3.468 after the first epoch to 3.419 after the second epoch and then became flat through epoch six (3.419, 3.419, 3.419, and 3.419). The possible explanation is that the learning rate was unsuitable for the larger model, however, this cannot be established conclusively without additional hyperparameter experiments.

3. **Table 3. Stability (5-fold post-hoc analysis)**

Model	ROUGE-1 (mean±std)	sacreBLEU (mean±std)	SBERT STS (mean±std)

T5-small (fine-tuned)	0.259 ± 0.011	5.47 ± 0.46	0.636 ± 0.021
T5-base (fine-tuned)	0.172 ± 0.005	1.43 ± 0.17	0.556 ± 0.013

4. **Table 4. Numerical Grounding**

Model	Mean numeric support rate	Examples with generated numbers
T5-small (fine-tuned)	0.421	686 / 795
T5-base (fine-tuned)	0.376	684 / 795

5. **Table 5. BART K-Fold Cross Validation Summary**

Model	RougeL	scareBLEU	Bert_F1	METEOR
BART	0.09865 (+/- 0.00258)	0.01539 (+/- 0.00237)	0.84309 (+/- 0.00142)	0.05658 (+/- 0.00153)

References

Banerjee, S. and Lavie, A. (2005). METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization.

Bahdanau, D., Cho, K., and Bengio, Y. (2014). Neural machine translation by jointly learning to align and translate. arXiv preprint arXiv:1409.0473.

Chapman, M. et al. (2021). Towards generating financial reports from table data using transformers. In Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing.

Lin, C.-Y. (2004). ROUGE: A package for automatic evaluation of summaries. In Proceedings of the ACL Workshop on Text Summarization Branches Out.

Min, S. et al. (2023). FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing.

Papineni, K., Roukos, S., Ward, T., and Zhu, W.-J. (2002). Bleu: a method for automatic evaluation of machine translation. In Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics.

Post, M. (2018). A call for clarity in reporting BLEU scores. In Proceedings of the Third Conference on Machine Translation.

Reiter, E. and Dale, R. (2000). Building Natural Language Generation Systems. Cambridge University Press.

See, A., Liu, P. J., and Manning, C. D. (2017). Get to the point: Summarization with pointer-generator networks. In Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics.

Vaswani, A. et al. (2017). Attention is all you need. Advances in Neural Information Processing Systems.

Wiseman, S., Shieber, S. M., and Rush, A. M. (2017). Challenges in data-to-document generation. In Proceedings of the 2017 Conference on Empirical Methods in Natural Language Processing.

Wiseman, S., Shieber, S. M., and Rush, A. M. (2018). Learning neural templates for data-to-document generation. In Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing.

Yang, Y. et al. (2024). DataTales: A benchmark for financial data-to-text generation. arXiv preprint.

Yang, Y. et al. (2025). KAHAN: Knowledge-Augmented Hierarchical Analysis and Narration for financial report generation. arXiv preprint arXiv:2509.17037.

Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., and Artzi, Y. (2020). BERTScore: Evaluating text generation with BERT. In Proceedings of the 8th International Conference on Learning Representations.